b) Two matrices are row equivalent if they have the same number of rows.

$\boxed{\text{Soln:}}$ In the second last paragraph on Page 6, we see the sufficient condition for two matrices to be $\boxed{\text{row equivalent}}$ are if there exists a sequence of elementary row operations that transforms one matrix into the other. Having the same no. of rows is not a sufficient condition for two matrices to be row equivalent. Hence, the given statement is False.

c) An inconsistent system has more than one solution.

Ans: On page 4, just above the section "Matrix Notation", we see that a system is inconsistent if it has no solution. ∴ The given statement is False.

d) Two linear systems are equivalent if they have the some solution set. (Wrong)

Ans: Same solution as (a).

25) Find an equation involving g, h and K that makes this augmented matrix correspond to a consistent system:

$$\begin{bmatrix} 1 & -4 & 7 & g \\ 0 & 3 & -5 & h \\ -2 & 5 & -9 & K \end{bmatrix}$$

Row 3 = Row 3 + 2 × Row 1

$$\begin{bmatrix} 1 & -4 & 7 & g \\ 0 & 3 & -5 & h \\ 0 & 6 & -11 & g+2K \end{bmatrix}$$

Row 3 = Row 3 − 2 × Row 2

$$\begin{bmatrix} 1 & -4 & 7 & g \\ 0 & 3 & -5 & h \\ 0 & 0 & -1 & g+2K-2h \end{bmatrix}$$

Row 3 = Row 3 × −1 ⟶

$$\begin{bmatrix} 1 & -4 & 7 & g \\ 0 & 3 & -5 & h \\ 0 & 0 & 1 & 2h-2K-g \end{bmatrix}$$

Row 2 = Row 2 + 5 × Row 3

$$\begin{bmatrix} 1 & -4 & 7 & g \\ 0 & 3 & 0 & 11h-2K-g \\ 0 & 0 & 1 & 2h-2K-g \end{bmatrix}$$

Row 1 = Row 1 − 7 × Row 3

$$\begin{bmatrix} 1 & -4 & 0 & 8g+2K-2h \\ 0 & 3 & 0 & 11h-2K-g \\ 0 & 0 & 1 & 2h-2K-g \end{bmatrix}$$